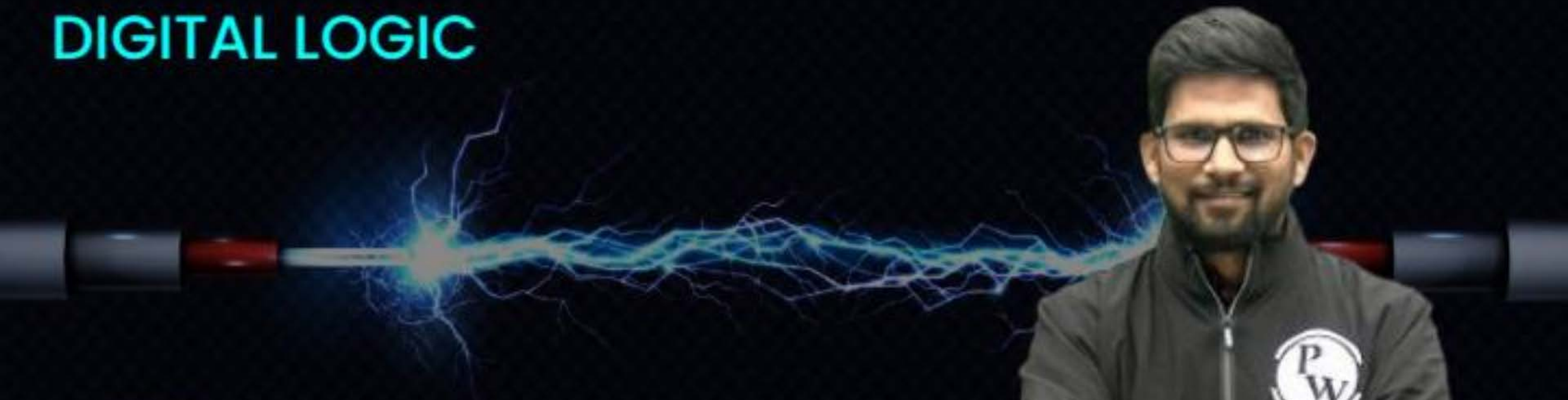
COMPUTER SCIENCE & IT

DIGITAL LOGIC





Lecture No: 05

Miscellaneous Topics



By- Chandan Gupta Sir

Recap of Previous Lecture



- Number System
-
-



- Prime implicants, EPI
- Question Discussion

Implicants

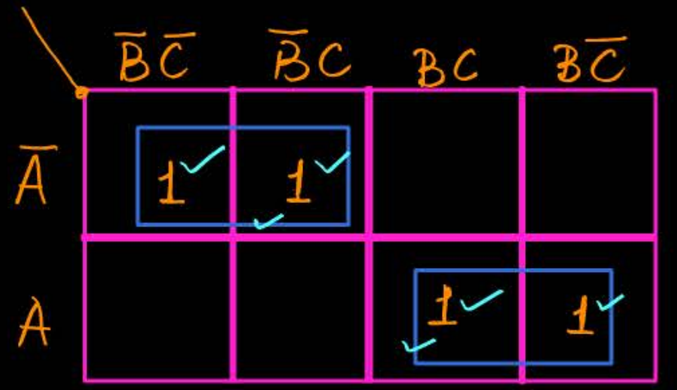
All single 1's, all possible pairs, all possible quads, all possible octates, all possible groups are called as implicants.

Prime Implicant:

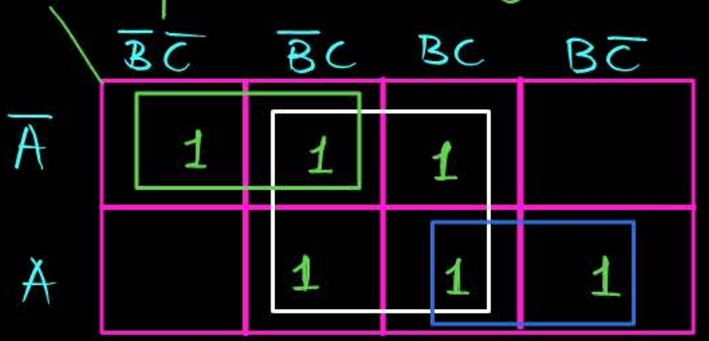
PI → The groups [implicants] which can not be converted in bigger group are called as prime implicants.

- EPI: The PIs which has atleast single 1' which can be covered only by this PI are called as EPI.

Essential PI.



implicants → 6

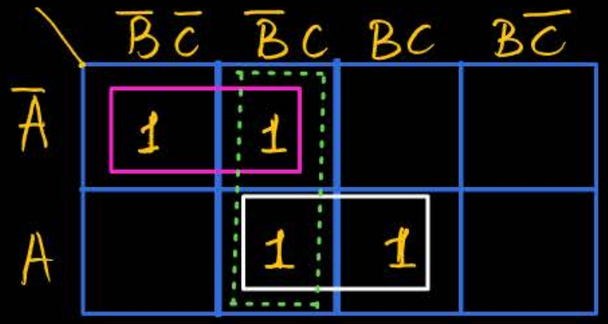


→ 6 pairs, 6 – single 1's & one quad

↓ implicants.

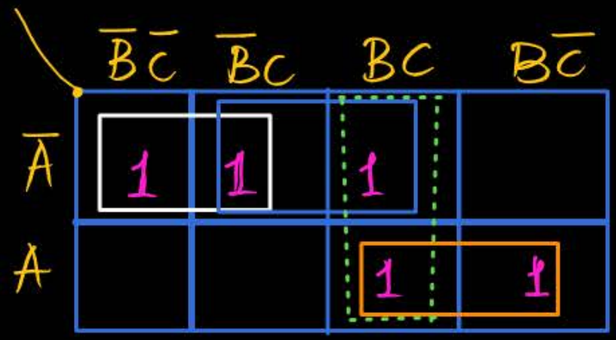
PI → $\bar{A}\bar{B}$, AB, C

EPI → $\bar{A}\bar{B}$, AB, C



PI → 3 → $\bar{A}\bar{B}$, AC, $\bar{B}C$

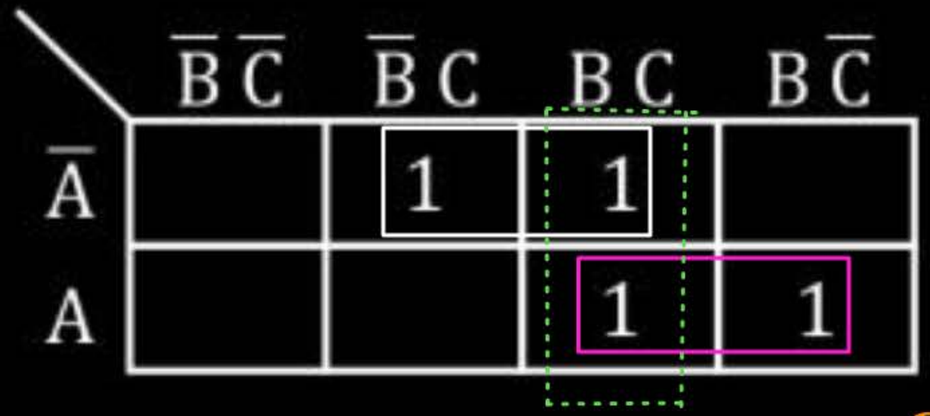
EPI → $\bar{A}\bar{B}$, AC



PI → $\bar{A}\bar{B}$, $\bar{A}C$, AB, BC

EPI → $\bar{A}\bar{B}$, AB

Examples :



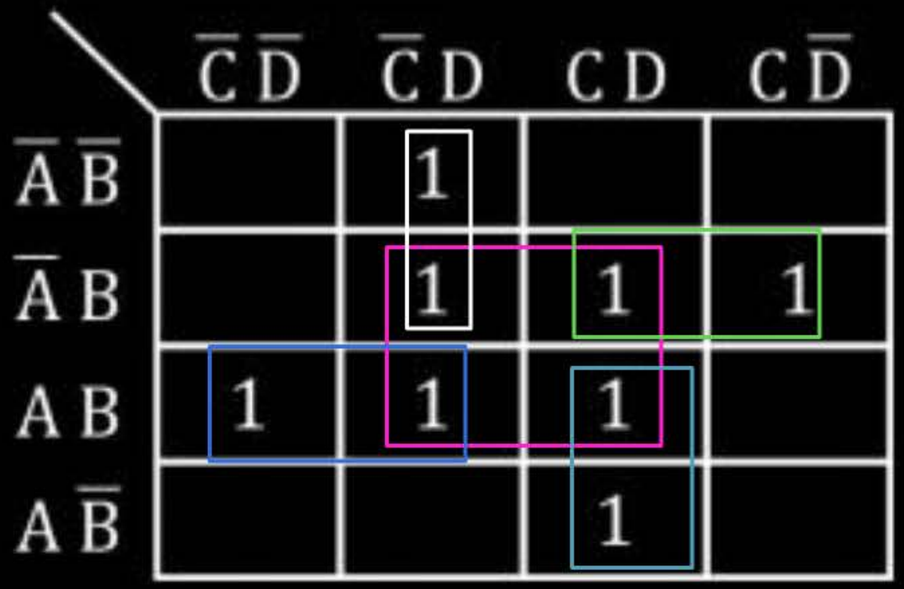
Prime Implicants → $\bar{A}C$, AB, BC → RPI

$\bar{A}C$, AB → NRPI

Essential Prime Implicants → $\bar{A}C$, AB

Non-Essential Prime Implicants → BC

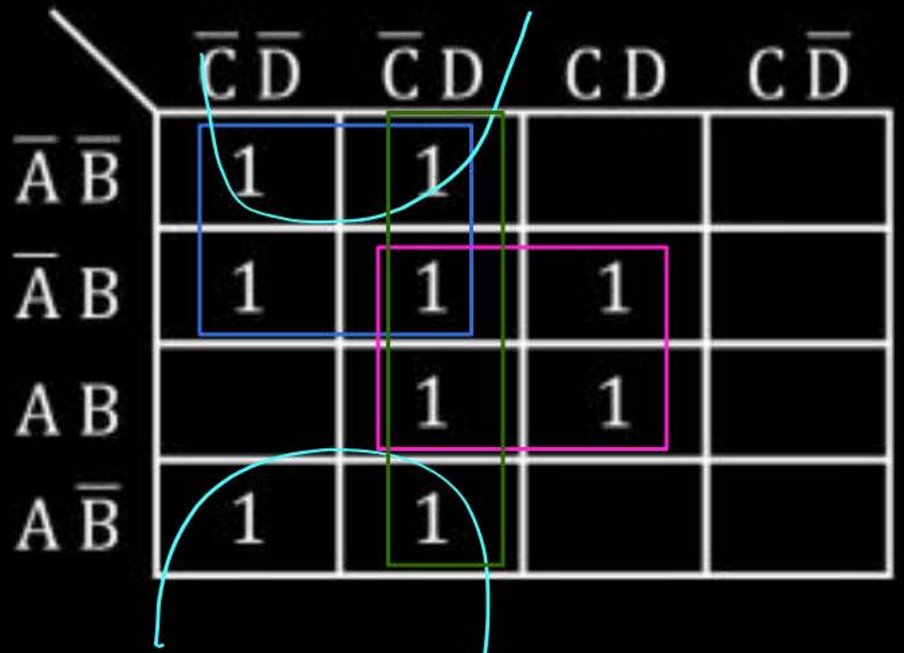
Examples :



Prime Implicants → $\bar{A}\bar{C}D$, $\bar{A}BC$, $AB\bar{C}$, ACD, BD → RPI

Essential Prime Implicants → $\bar{A}\bar{C}D$, $\bar{A}BC$, $AB\bar{C}$, ACD

Examples :



Prime Implicants → $\bar{A}\bar{C}, \bar{B}\bar{C}, BD, \bar{C}D$

Essential Prime Implicants → $\bar{A}\bar{C}, \bar{B}\bar{C}, BD$

[NAT]

$2^{-1}\,2^{-2}\,2^{-3}\,2^{-4}$

If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, the decimal value of x + y is 3.

[GATE-2021-CS: 1M]

$= (0.5 + 0.25 + 0.0625)_{10}$

$= (0.8125)_{10}$

$x = 1$

$y = 2$

[NAT]

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b. The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then b = 8.

[GATE-2017-CS: 1M]

$x^2 - (\alpha+\beta)x + \alpha\cdot\beta = 0$

$\alpha = (5)_b \longrightarrow 5$

$\beta = (6)_b \longrightarrow 6$

$\alpha+\beta = (13)_b \longrightarrow (13)_8$

$[x^2 - (b+3)x + (3b+6)]_{10} = 0$

$x = (6)_b = (6)_{10}$

$36 - (b+3)6 + 3b + 6 = 0$

$3b = 24$

$b = 8$

[NAT]

Consider the euqation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solution is 5.

[GATE-2015-CS: 2M]

$x \geq 5$ & $0 \leq y \leq 7$

$$(4x+3)_{10} = (8y+3)_{10}$$

$$\boxed{x = 2y}$$

$3 \leq y \leq 7 \longrightarrow 6 \leq x \leq 14$

(3,6), (4,8), (5,10), (6,12), (7,14)

[NAT]

The base (or radix) of the number system such that the following equation holds is 5. 312/20 = 13.1

[GATE-2014-CS: 1M]

$$\frac{(312)_b}{(20)_b} = (13.1)_b \longrightarrow b \geq 4$$

$$(312)_b = (13.1)_b \times (20)_b$$

$$(3b^2+b+2)_{10} = (b+3+b^{-1})_{10}\,(2b)$$

$$3b^2+b+2 = 2b^2+6b+2$$

$$b^2-5b=0$$

$$b(b-5)=0 \longrightarrow b = 0\ (\times),\ 5\ (\checkmark)$$

[NAT]

Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is 3.

[GATE-2014-CS: 1M]

$5^2 5^1 5^0$ $\quad$ $y^1 y^0$

$(5^2 \times 1 + 5 \times 2 + 5^0 \times 3)_{10} = (y^1 \times x + 8)_{10}$

$38 = xy + 8$

$xy = 30$

$y \geq 9$ & $y \geq (x+1)$

$0 \leq x \leq y-1$

1×30, 2×15, 3×10, 5×6 ✗

[MCQ]

Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r} = 11_r$ is/are [GATE-2008-CS: 1M]

A. Decimal 10

B. Decimal 11

C. Decimal 10 and 11

D. Any value > 2



[MCQ]

Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as A=01010 and B = 11010 . Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow? [GATE-2024-CS: 1M]

$A = (01010)_2 = +(10)_{10}$

$B = (11010)_2 = (-6)_{10}$

HW

$-(16)$ to $+(15)$

A. ✗ A + B $(4)_{10}$

B. ✓ A − B 16 → overflow.

C. ✗ B − A $(-16)_{10}$ ✓

D. 2 * B $(-12)_{10}$

[MCQ]

In 16-bit 2's complement representation, the decimal number -28 is:

[GATE-2019-CS: 1M]

A. 1111 1111 0001 1100

B. 0000 0000 1110 0100

C. 1111 1111 1110 0100

D. 1000 0000 1110 0100

$1111111111100100)_2 = (-28)_{10}$

[NAT]

Two numbers are chosen independently and uniformly at random from the set {1, 2, ... , 13}. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

[GATE-2019-CS: 1M]

H/W

[MCQ]

Consider Z = X - Y, where X, Y and Z are all in sign-magnitude form. X and Y are each represented in n bits . To avoid overflow , the representation of Z would require a minimum of : [GATE-2019-CS: 1M]

3 bit → — (3) to (3)

X-Y → (−6 to 6)

4 bit → (−7 to +7)

HW

A. n bits

B. n-1 bits

C. n+1 bits

D. n+2 bits

[MCQ]

Consider the unsigned 8-bit fixed point binary number representation below,

$$b_7\ b_6\ b_5\ b_4\ b_3 \cdot b_2\ b_1\ b_0$$

where the position of the binary point is between b_3 and b_2 . Assume b_7 is the most significant bit. Some of the decimal numbers listed below cannot be represented exactly in the above representation:

(i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001

Which one of the following statements is true? [GATE-2018-CS: 1M]

A. None of (i), (ii), (iii), (iv) can be exactly represented

B. Only (ii) cannot be exactly represented

C. Only (iii) and (iv) cannot be exactly represented

D. Only (i) and (ii) cannot be exactly represented

H.W

[MCQ]

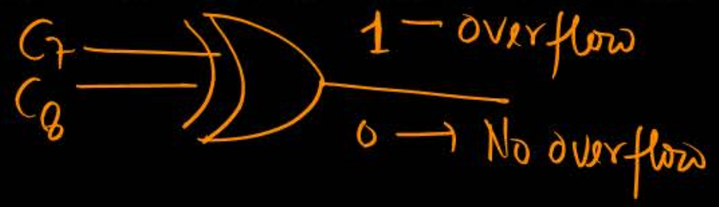
When two 8-bit numbers $A_7 \ldots A_0$ and $B_7 \ldots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a ripple-carry adder, the sum bits obtained are $S_7 \ldots S_0$ and the carry bits are $C_7 \ldots C_0$. An overflow is said to have occurred if [GATE-2017-CS: 1M]

A. the carry bit C_7 is 1

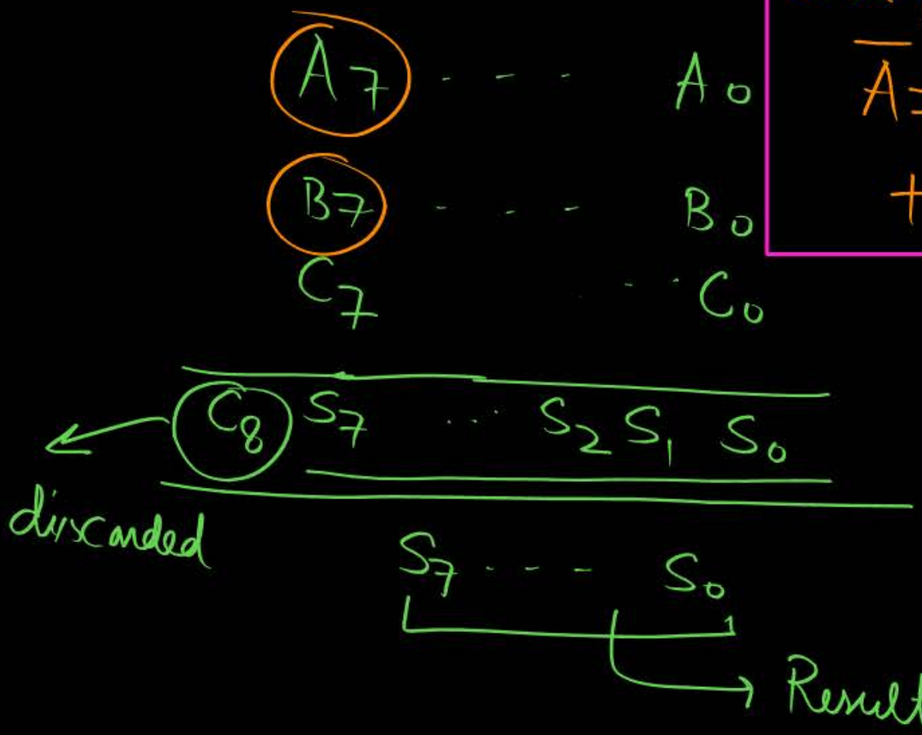
B. all the carry bits (C_7, ... ,C_0) are 1

C. $(A_7 \cdot B_7 \cdot \overline{S_7} + \overline{A_7} \cdot \overline{B_7} \cdot S_7)$ is 1.

D. $(A_0 \cdot B_0 \cdot \overline{S_0} + \overline{A_0} \cdot \overline{B_0} \cdot S_0)$ is 1.



Overflow =
$\overline{A_7}\,\overline{B_7} \cdot S_7 + A_7 B_7 \overline{S_7}$



[NAT]

Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then X-Y is _____. [GATE-2016-CS: 1M]

H.W

[NAT]

The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is___.

[GATE-2016-CS: 1M]

$= (10101)_2$

H.W

[MCQ]

The smallest integer that can be represented by an 8-bit number in 2's complement form is

[GATE-2013-CS: 1M]

A. –256

B. –128

C. –127

D. 0

H.W

[MCQ]

P is a 16-bit signed integer. The 2's complement representation of P is $(F87B)_{16}$. The 2's complement representation of 8×P is [GATE-2010-CS: 1M]

A. $(C3D8)_{16}$

B. $(187B)_{16}$

C. $(F878)_{16}$

D. $(987B)_{16}$

$P = (1111\ 1000\ 0111\ 1011)_2$

8 X P

$2^3 \times P = (1111\ 1000\ 0111\ 1011\ 000)_2$

$= (11000\ 0111\ 1011\ 000)_2$

$= (C3D8)_{16}$

$= (FC3D8)_{16}$

[MCQ]

Which of the following is/are EQUAL to 224 in radix-5 (i.e., base-5) notation?

[GATE-2014-CS: 1M]

A. 64 in radix-10

B. 100 in radix-8

C. 121 in radix-7

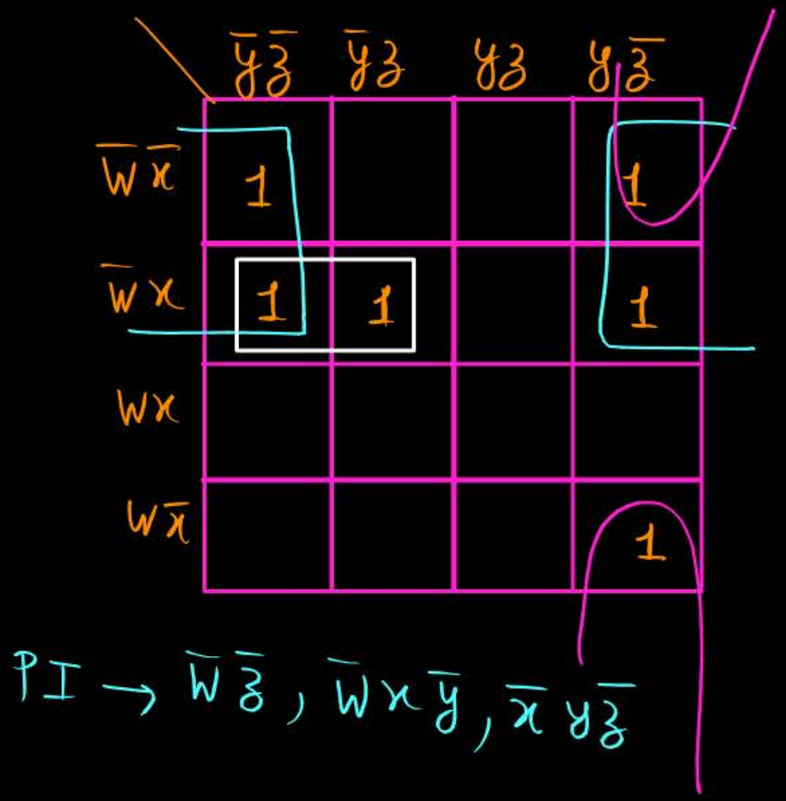
D. 50 in radix-16

H.W.

[NAT]

The total number of prime implicants of the function $f(w,x,y,z) = \Sigma(0, 2, 4, 5, 6, 10)$ is 3.

[GATE-2022-CS: 1M]



[NAT]

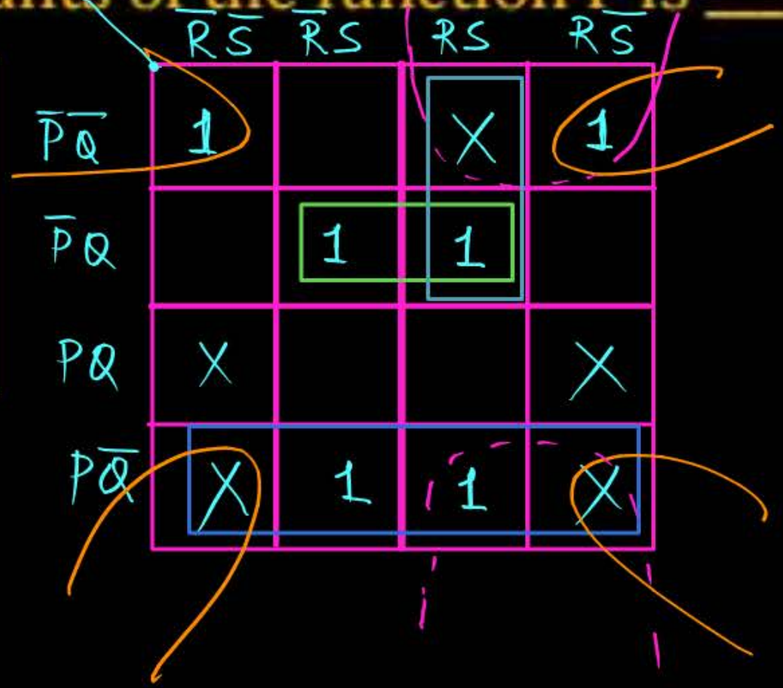
Consider the minterm list form of a Boolean function F given below.

$F(P, Q, R, S) = \Sigma m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is 3. [GATE-2018-CS: 2M]

PI → 5 → $P\bar{Q}$, $\bar{Q}\bar{S}$, $\bar{P}QS$, $\bar{Q}R$, $\bar{P}RS$

EPI → $P\bar{Q}$, $\bar{Q}\bar{S}$, $\bar{P}QS$



$PI_x \longrightarrow 5$



Topic : 2 Min Summary

- PI & EPI
- Question Discussion.

Thank you

GW

Soldiers !

